

WEB-BASED ML MODEL EVALUATOR

TEST PLAN DOCUMENT

Test cases based on the provided Use Case Flow

UC-01 Register

Module: Analyst Registration | Test coverage: 3 Unit + 3 Integration + 2 System

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
01_UT_01	Analyst Registration	Validate required registration fields	Registration page/module is available.	Open registration; leave required fields blank; submit.	Blank required fields	System identifies invalid/incomplete information and requests correction.	Not Executed	Not Executed
01_UT_02	Analyst Registration	Validate registration details	Registration form is available.	Enter valid and invalid values in the registration fields; submit.	Valid details; invalid/incomplete details	Valid information passes validation; invalid/incomplete information is rejected with an error.	Not Executed	Not Executed
01_UT_03	Analyst Registration	Check duplicate account	An account with the specified details exists.	Enter the existing account details; submit.	Existing username/email/details	Registration is rejected and an appropriate duplicate-account error is shown.	Not Executed	Not Executed
01_IT_01	Analyst Registration	Create Analyst account after validation	Valid unique registration details are entered.	Submit valid details; allow validation and account creation.	Unique valid Analyst details	System creates the Analyst account and proceeds to successful creation/dashboard flow.	Not Executed	Not Executed
01_IT_02	Analyst Registration	Prevent Admin creation through registration	Registration function is available.	Attempt to register using registration details intended to create an Admin account.	Admin role request	System does not create an Admin account; Admin access can only be granted later by an existing Admin.	Not Executed	Not Executed
01_IT_03	Analyst Registration	Open Analyst dashboard after registration	Valid unique registration details are available.	Register successfully; observe the next screen.	Unique valid details	System informs the Analyst of successful creation and opens the Analyst dashboard.	Not Executed	Not Executed
01_ST_01	Analyst Registration	Successful Analyst registration end-to-end	Registration page is accessible.	Enter valid unique details; submit; verify account and dashboard.	Valid unique Analyst details	Registration succeeds, account is created, success information is displayed, and Analyst dashboard opens.	Not Executed	Not Executed
01_ST_02	Analyst Registration	Invalid/duplicate registration handling	Registration page is accessible.	Submit incomplete/invalid details and then duplicate details.	Invalid details; existing account details	System shows readable error messages and does not create an invalid or duplicate account.	Not Executed	Not Executed

UC-02 Login

Module: Login | Test coverage: 3 Unit + 3 Integration + 2 System

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
02_UT_01	Login	Validate login input fields	Login page is available.	Enter blank or malformed credentials and submit.	Blank/invalid username or password	System validates input and prompts the user to correct invalid details.	Not Executed	Not Executed
02_UT_02	Login	Authenticate valid credentials	A valid Analyst/Admin account exists.	Enter correct username and password; submit.	Valid username and password	Credentials are accepted and authentication succeeds.	Not Executed	Not Executed
02_UT_03	Login	Reject invalid credentials	Login page is available.	Enter incorrect username/password; submit.	Incorrect credentials	System displays an error and prompts the user to enter credentials again.	Not Executed	Not Executed
02_IT_01	Login	Route Analyst to Analyst dashboard	Valid Analyst account exists.	Log in using Analyst credentials.	Analyst credentials	System authenticates the user and navigates to the Analyst-specific dashboard.	Not Executed	Not Executed

02_IT_02	Login	Route Admin to Admin dashboard	Valid Admin account exists.	Log in using Admin credentials.	Admin credentials	System authenticates the user and navigates to the Admin-specific dashboard.	Not Executed	Not Executed
02_IT_03	Login	Deny blocked account	A user account is blocked.	Attempt login using the blocked account credentials.	Blocked account credentials	System denies login.	Not Executed	Not Executed
02_ST_01	Login	Successful Analyst login	Valid Analyst account exists.	Navigate to login; enter valid credentials; submit.	Valid Analyst credentials	User is authenticated and receives the Analyst dashboard.	Not Executed	Not Executed
02_ST_02	Login	Login failure and blocked-account flow	Login page is available; invalid and blocked accounts can be tested.	Attempt invalid login and blocked-account login.	Incorrect credentials; blocked account	Invalid login is rejected with an error; blocked account is denied access.	Not Executed	Not Executed

UC-03 Upload Dataset

Module: Dataset Upload | Test coverage: 3 Unit + 3 Integration + 2 System

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
03_UT_01	Dataset Upload	Validate dataset file format	Dataset upload screen is available.	Select a supported dataset file and an unsupported file; observe validation.	Supported file; unsupported file	Supported format is accepted; invalid format is rejected with a readable explanation.	Not Executed	Not Executed
03_UT_02	Dataset Upload	Validate dataset schema	Dataset upload screen is available.	Select a dataset with valid schema and one with invalid/corrupt schema.	Valid dataset; malformed dataset	Valid schema passes; invalid/corrupt dataset is rejected with an explanation.	Not Executed	Not Executed
03_UT_03	Dataset Upload	Validate target column selection	A valid dataset is available.	Open target-column selection and choose a target column.	Valid target column; missing/invalid target	System accepts a valid target and prevents an invalid/missing target selection.	Not Executed	Not Executed
03_IT_01	Dataset Upload	Load accepted dataset	A valid dataset file is available.	Open upload screen; select file; choose target column; submit.	Valid dataset + target column	System validates the file/schema and loads the dataset for evaluation.	Not Executed	Not Executed
03_IT_02	Dataset Upload	Reject corrupt dataset	Upload screen is available.	Select a corrupt/invalid dataset file.	Corrupt dataset file	System rejects the file and provides a readable reason such as inability to load the file.	Not Executed	Not Executed
03_IT_03	Dataset Upload	Make dataset available for evaluation	Dataset has been accepted and loaded.	Complete upload and observe the evaluation workflow.	Valid dataset	Loaded data becomes available to the Analyst for model evaluation.	Not Executed	Not Executed
03_ST_01	Dataset Upload	Successful dataset upload	Analyst is authenticated.	Open upload screen; choose a valid dataset; choose target column; submit.	Valid dataset + valid target	Dataset is accepted, loaded, and made available for evaluation.	Not Executed	Not Executed
03_ST_02	Dataset Upload	Invalid dataset upload handling	Analyst is authenticated.	Upload unsupported/corrupt or invalid-schema data.	Invalid/corrupt dataset	System rejects the file and displays a readable explanation without loading it for evaluation.	Not Executed	Not Executed

UC-04 Upload Model

Module: Model Upload | Test coverage: 3 Unit + 3 Integration + 2 System

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
04_UT_01	Model Upload	Validate model file format	Model upload function is accessible.	Select a supported trained-model file and an inappropriate model file.	Valid model file; invalid model file	Supported model file is accepted; inappropriate/invalid model file is rejected.	Not Executed	Not Executed
04_UT_02	Model Upload	Validate model file selection	Model upload screen is available.	Attempt to continue without selecting a model file.	No file selected	System prevents upload and requests a model file.	Not Executed	Not Executed
04_UT_03	Model Upload	Validate model availability	A valid trained model	Select and upload the model file.	Valid trained model	Model passes validation and	Not Executed	Not

		after upload	file is available.			is made available.		Executed
04_IT_01	Model Upload	Upload trained model successfully	Model upload function is available.	Select trained model; submit; observe validation.	Valid trained model file	File validation completes and the model becomes available for evaluation.	Not Executed	Not Executed
04_IT_02	Model Upload	Check model-dataset compatibility	A dataset has already been uploaded.	Upload a model whose expected feature set is compatible/incompatible with dataset schema.	Compatible model; incompatible model	Compatible model is accepted; incompatible model is rejected with an explanation.	Not Executed	Not Executed
04_IT_03	Model Upload	Reject invalid model file	Model upload function is available.	Select an invalid or inappropriate model file.	Invalid model format/file	System immediately rejects the model and explains the problem.	Not Executed	Not Executed
04_ST_01	Model Upload	Successful model upload	Analyst is authenticated.	Access model upload; select trained model; submit.	Valid trained model	Model is validated and made available to the Analyst.	Not Executed	Not Executed
04_ST_02	Model Upload	Model upload with incompatible dataset	Dataset was uploaded previously.	Upload an incompatible model.	Model with incompatible feature set	System rejects the model and explains the incompatibility; model is not made available for evaluation.	Not Executed	Not Executed

UC-05 Select Task Type

Module: Task Type Selection | Test coverage: 3 Unit + 3 Integration + 2 System

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
05_UT_01	Task Type Selection	Display task options	Analyst is on task selection screen.	Open task selection.	None	Classification and regression options are displayed.	Not Executed	Not Executed
05_UT_02	Task Type Selection	Select classification	Task selection screen is available.	Choose classification.	Classification	Classification is selected.	Not Executed	Not Executed
05_UT_03	Task Type Selection	Select regression	Task selection screen is available.	Choose regression.	Regression	Regression is selected.	Not Executed	Not Executed
05_IT_01	Task Type Selection	Verify classification compatibility	Uploaded dataset and model are available.	Select classification; submit/continue.	Classification task + classification-compatible data/model	System verifies the selection and activates appropriate metrics/visualizations.	Not Executed	Not Executed
05_IT_02	Task Type Selection	Verify regression compatibility	Uploaded dataset and model are available.	Select regression; submit/continue.	Regression task + regression-compatible data/model	System verifies the selection and activates appropriate metrics/visualizations.	Not Executed	Not Executed
05_IT_03	Task Type Selection	Reject mismatched task	Uploaded model/dataset pair is available.	Select classification for regression data/model or vice versa.	Mismatched task type	System denies evaluation and shows an indication/reasoning for the mismatch.	Not Executed	Not Executed
05_ST_01	Task Type Selection	Successful classification task selection	Valid uploaded model and dataset are available.	Open task selection; choose classification; continue.	Classification	Classification is accepted and relevant evaluation options become available.	Not Executed	Not Executed
05_ST_02	Task Type Selection	Mismatched task selection	Uploaded model/dataset pair is available.	Choose a task inconsistent with the uploaded data/model.	Wrong task type	System prevents evaluation and displays a suitable explanation.	Not Executed	Not Executed

UC-06 Evaluate Model

Module: Model Evaluation | Test coverage: 3 Unit + 3 Integration + 2 System

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
06_UT_01	Model Evaluation	Validate evaluation pair	Evaluation screen is available.	Open evaluation for an uploaded model/dataset pair.	Valid model + dataset	System verifies that the model and dataset can be used together.	Not Executed	Not Executed
06_UT_02	Model Evaluation	Generate prediction result	Compatible model and dataset are available.	Run evaluation.	Compatible evaluation pair	System runs the model against the evaluation dataset and produces prediction	Not Executed	Not Executed

						results.		
06_UT_03	Model Evaluation	Calculate task metrics	Prediction results are available.	Complete metric calculation for selected task.	Classification/regression prediction results	System calculates metrics relevant to the stated task type.	Not Executed	Not Executed
06_IT_01	Model Evaluation	Evaluate uploaded model against dataset	Compatible model and dataset are uploaded.	Start evaluation; wait for prediction and metric calculation.	Model + dataset + task type	System validates the pair, runs prediction, and calculates relevant metrics.	Not Executed	Not Executed
06_IT_02	Model Evaluation	Render evaluation visualization	Evaluation has produced the required values.	Complete evaluation and observe visualization rendering.	Classification or regression evaluation	System renders visualization appropriate to the selected task.	Not Executed	Not Executed
06_IT_03	Model Evaluation	Store completed evaluation metadata	Evaluation completes successfully.	Finish evaluation and inspect stored result/history.	Model ID, dataset ID, task type, timestamp	Evaluation result, visualization and specified metadata are recorded alongside the Analyst account.	Not Executed	Not Executed
06_ST_01	Model Evaluation	Successful end-to-end evaluation	Analyst is authenticated; model/dataset are compatible.	Open evaluation; run model; wait for metrics and visualization.	Valid model + dataset + task	System completes evaluation, stores results, and shows results to Analyst.	Not Executed	Not Executed
06_ST_02	Model Evaluation	Evaluation exception handling	Evaluation can be initiated with an invalid/problematic pair.	Run evaluation and trigger/encounter a prediction or metric exception.	Problematic evaluation input	System terminates evaluation without storing partial results and displays a message explaining why it terminated.	Not Executed	Not Executed

UC-07 View Metrics

Module: Metrics Viewing | Test coverage: 3 Unit + 3 Integration + 2 System

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
07_UT_01	Metrics Viewing	Display metric values	A completed analysis exists.	Open completed analysis.	Completed evaluation	Metric values for the task are displayed.	Not Executed	Not Executed
07_UT_02	Metrics Viewing	Handle missing analysis	No completed analysis is available.	Attempt to open metrics.	Unavailable analysis	System displays a suitable message indicating analysis is not available.	Not Executed	Not Executed
07_UT_03	Metrics Viewing	Display task-relevant metrics	Completed analysis contains metric values.	Open metrics for the selected task.	Classification/regression metrics	System shows the metric values associated with that task.	Not Executed	Not Executed
07_IT_01	Metrics Viewing	Retrieve metrics from evaluation	A stored completed evaluation exists.	Open the evaluation's metrics view.	Stored evaluation ID	System retrieves the corresponding metric values.	Not Executed	Not Executed
07_IT_02	Metrics Viewing	Show metrics with analysis	Completed analysis and metrics exist.	Open analysis and navigate to metrics.	Completed evaluation	Metrics are displayed with the selected completed analysis.	Not Executed	Not Executed
07_IT_03	Metrics Viewing	Handle unavailable stored metrics	Evaluation exists but required metric information is unavailable.	Attempt to open metrics.	Evaluation with unavailable metrics	System does not show misleading values and displays a suitable availability message.	Not Executed	Not Executed
07_ST_01	Metrics Viewing	View metrics for completed evaluation	Analyst has a completed evaluation.	Open completed analysis and view metrics.	Completed evaluation	Correct metric values are shown to the Analyst.	Not Executed	Not Executed
07_ST_02	Metrics Viewing	View metrics when analysis is unavailable	No completed analysis is available.	Attempt to view metrics.	No analysis	System shows an appropriate message instead of metric values.	Not Executed	Not Executed

UC-08 View Visualizations

Module: Visualization Viewing | Test coverage: 3 Unit + 3 Integration + 2 System

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
08_UT_01	Visualization Viewing	Render classification confusion matrix	Completed classification evaluation exists.	Open finalized classification evaluation.	Classification result	Confusion matrix is plotted when computed.	Not Executed	Not Executed

08_UT_02	Visualization Viewing	Render classification ROC curve when probabilities exist	Classification model produced probabilities.	Open finalized evaluation.	Classification + predicted probabilities	ROC curve is plotted.	Not Executed	Not Executed
08_UT_03	Visualization Viewing	Render regression R graph	Completed regression evaluation exists.	Open finalized regression evaluation.	Regression result	R graph is plotted.	Not Executed	Not Executed
08_IT_01	Visualization Viewing	Display visualizations for completed evaluation	Evaluation is finalized and visualizations were computed.	Open the evaluation visualization view.	Completed evaluation	System plots the visualizations defined for the task.	Not Executed	Not Executed
08_IT_02	Visualization Viewing	Handle missing probability for ROC	Classification model did not predict probabilities.	Open finalized classification evaluation.	Classification without probabilities	System does not plot ROC curve and displays only the visualizations that were computed.	Not Executed	Not Executed
08_IT_03	Visualization Viewing	Match visualization to task type	Completed classification/regression evaluations exist.	Open each finalized evaluation.	Classification; regression	Classification uses applicable classification visualizations; regression uses applicable regression visualizations.	Not Executed	Not Executed
08_ST_01	Visualization Viewing	View all available classification visualizations	Completed classification evaluation with required outputs exists.	Open finalized classification evaluation.	Classification + computed outputs	Available classification visualizations are displayed correctly.	Not Executed	Not Executed
08_ST_02	Visualization Viewing	View visualization when ROC data is unavailable	Classification evaluation lacks predicted probabilities.	Open finalized evaluation.	No probability output	ROC curve is omitted and only possible computed visualizations are displayed.	Not Executed	Not Executed

UC-09 Compare Models

Module: Model Comparison | Test coverage: 3 Unit + 3 Integration + 2 System

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
09_UT_01	Model Comparison	Select two completed evaluations	At least two completed evaluations exist.	Open comparison and select two evaluations.	Two completed evaluations	Selected evaluations are accepted for comparison.	Not Executed	Not Executed
09_UT_02	Model Comparison	Display metrics side-by-side	Two comparable evaluations are selected.	Run comparison.	Two same-task evaluations	Metrics are presented side-by-side.	Not Executed	Not Executed
09_UT_03	Model Comparison	Reject empty comparison selection	Comparison page is available.	Attempt comparison without selecting required evaluations.	No evaluation selected	System prevents comparison or requests valid selections.	Not Executed	Not Executed
09_IT_01	Model Comparison	Verify same task type	Completed evaluations of different task types exist.	Select evaluations with different task types.	Classification + regression	System detects task-type mismatch and warns the Analyst.	Not Executed	Not Executed
09_IT_02	Model Comparison	Compare same-task evaluations	Completed evaluations of the same task type exist.	Select two or more same-task evaluations.	Two classification or two regression evaluations	System enables comparison and presents their respective metrics side-by-side.	Not Executed	Not Executed
09_IT_03	Model Comparison	Retrieve metrics for selected evaluations	Selected evaluations have stored metrics.	Start comparison.	Evaluation IDs with metrics	System retrieves and displays metrics for each selected evaluation.	Not Executed	Not Executed
09_ST_01	Model Comparison	Successful model comparison	Two or more completed same-task evaluations exist.	Open comparison; select evaluations; compare.	Same-task evaluations	Corresponding metrics are displayed side-by-side.	Not Executed	Not Executed
09_ST_02	Model Comparison	Mismatched model comparison	Completed classification and regression evaluations exist.	Select evaluations of different task types.	Classification + regression evaluations	System warns about the condition and does not enable comparison.	Not Executed	Not Executed

UC-10 View Evaluation History

Module: Evaluation History | Test coverage: 3 Unit + 3 Integration + 2 System

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
10_UT_01	Evaluation History	Open evaluation history	Analyst is	Navigate to evaluation history.	Authenticated	Evaluation history page	Not Executed	Not

			authenticated.		Analyst	opens.		Executed
10_UT_02	Evaluation History	List saved evaluations	Analyst has saved evaluations.	Open history page.	One or more saved evaluations	System populates the Analyst's saved evaluation history.	Not Executed	Not Executed
10_UT_03	Evaluation History	Handle empty history	Analyst has no saved evaluations.	Open history page.	No saved evaluations	System displays 'no evaluation history found'.	Not Executed	Not Executed
10_IT_01	Evaluation History	Retrieve history for current Analyst	Analyst has saved evaluations.	Access history using the Analyst account.	Analyst account with saved evaluations	System retrieves evaluations belonging to the Analyst.	Not Executed	Not Executed
10_IT_02	Evaluation History	Open selected historical evaluation	History contains at least one saved evaluation.	Select an evaluation from history.	Saved evaluation	Selected evaluation opens/reopens successfully.	Not Executed	Not Executed
10_IT_03	Evaluation History	Prevent unrelated history from appearing	Multiple Analyst accounts exist with separate histories.	Log in as one Analyst and open history.	Current Analyst + other Analyst history	Only the current Analyst's saved evaluations are populated.	Not Executed	Not Executed
10_ST_01	Evaluation History	View and reopen saved evaluation	Analyst has at least one saved evaluation.	Open history; select an evaluation.	Saved evaluation	History is shown and the selected evaluation can be reopened.	Not Executed	Not Executed
10_ST_02	Evaluation History	View history with no saved evaluations	Analyst has no saved evaluation history.	Open history page.	Empty history	System displays 'no evaluation history found'.	Not Executed	Not Executed

UC-11 Generate Report

Module: Report Generation | Test coverage: 3 Unit + 3 Integration + 2 System

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
11_UT_01	Report Generation	Select one saved evaluation	At least one saved evaluation exists.	Open report function and select one evaluation.	One saved evaluation	Evaluation is accepted for report generation.	Not Executed	Not Executed
11_UT_02	Report Generation	Select multiple saved evaluations	Two or more saved evaluations exist.	Select two or more saved evaluations.	Multiple saved evaluations	Selected evaluations are accepted for aggregation/comparison.	Not Executed	Not Executed
11_UT_03	Report Generation	Validate missing evaluation selection	Report function is available.	Attempt to generate a report without a valid saved evaluation.	No/invalid selection	System does not generate an invalid report and prompts for valid selection.	Not Executed	Not Executed
11_IT_01	Report Generation	Aggregate model information	Selected saved evaluation exists.	Generate report.	Saved evaluation with model metadata	System retrieves model information for the report.	Not Executed	Not Executed
11_IT_02	Report Generation	Aggregate dataset, metrics and visualization information	Selected evaluation contains required stored information.	Generate report.	Evaluation with dataset/metrics/visualizations	System aggregates the available information.	Not Executed	Not Executed
11_IT_03	Report Generation	Generate report from selected evaluations	One or more valid saved evaluations are selected.	Request report generation.	Valid saved evaluation(s)	System outputs aggregated information as a report.	Not Executed	Not Executed
11_ST_01	Report Generation	Generate report for one evaluation	One saved evaluation is available.	Select it and generate report.	One valid saved evaluation	A report containing the relevant model, dataset, metrics and visualization information is generated.	Not Executed	Not Executed
11_ST_02	Report Generation	Report generation when evaluation is unavailable	A selected saved evaluation no longer exists.	Attempt to generate report using the unavailable evaluation.	Missing evaluation	System displays an error and does not generate the report.	Not Executed	Not Executed

UC-12 Manage User Accounts

Module: User Account Management | Test coverage: 3 Unit + 3 Integration + 2 System

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
12_UT_01	User Account Management	Display registered accounts	Admin is authenticated.	Open manage account section.	Admin account	System displays registered accounts and their statuses.	Not Executed	Not Executed
12_UT_02	User Account Management	Suspend active account	Admin is authenticated; target account is active.	Select active account and choose Suspend.	Active user account	Account status changes to suspended.	Not Executed	Not Executed
12_UT_03	User Account	Reactivate suspended	Admin is authenticated;	Select suspended account and	Suspended user	Account status changes to	Not Executed	Not

	Management	account	target account is suspended.	choose Reactivate.	account	active.		Executed
12_IT_01	User Account Management	Prevent Admin from manipulating personal account	Admin is authenticated.	Select the current Admin account and attempt account-status operation.	Current Admin account	System refuses the operation according to the scenario.	Not Executed	Not Executed
12_IT_02	User Account Management	Maintain at least one active Admin	More than one Admin exists.	Attempt to suspend/deactivate an Admin when it would leave no active administrator.	Last-active-Admin condition	System refuses the operation and informs the Admin.	Not Executed	Not Executed
12_IT_03	User Account Management	Record account-status operation in audit log	Admin can perform a valid account operation.	Suspend/reactivate an account; inspect audit information.	Admin ID + affected User ID + timestamp	System adds the activity to the audit log.	Not Executed	Not Executed
12_ST_01	User Account Management	Successful account suspension/reactivation	Admin is authenticated and target account has appropriate status.	Open account management; select user; suspend or reactivate.	Active/suspended target account	Selected account status is updated and operation is recorded.	Not Executed	Not Executed
12_ST_02	User Account Management	Invalid account-management operation	Admin is authenticated.	Attempt an operation that would manipulate personal account or remove the last active administrator.	Current Admin/last Admin condition	System refuses the operation and informs the Admin; account status is unchanged.	Not Executed	Not Executed

UC-13 Manage User Roles

Module: User Role Management | Test coverage: 3 Unit + 3 Integration + 2 System

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
13_UT_01	User Role Management	Display accounts and current roles	Admin is authenticated.	Open role management.	Admin account	System shows existing accounts with corresponding roles.	Not Executed	Not Executed
13_UT_02	User Role Management	Select Analyst role	Role management page is available.	Choose an account and select Analyst role.	Target account + Analyst role	Analyst role is selected for confirmation.	Not Executed	Not Executed
13_UT_03	User Role Management	Select Admin role	Role management page is available.	Choose an account and select Admin role.	Target account + Admin role	Admin role is selected for confirmation.	Not Executed	Not Executed
13_IT_01	User Role Management	Update account role	Admin is authenticated; target account exists.	Choose account; select new role; confirm.	Affected account ID + new role	System updates the account with the new role.	Not Executed	Not Executed
13_IT_02	User Role Management	Record role change in audit log	A role change is performed.	Change a user's role and inspect audit information.	Admin ID, affected account ID, old role, new role, timestamp	System logs the role-change entry with the specified information.	Not Executed	Not Executed
13_IT_03	User Role Management	Prevent Admin from modifying own role	Admin is authenticated.	Select own account and attempt to change its role.	Current Admin account	System denies the request.	Not Executed	Not Executed
13_ST_01	User Role Management	Successful user role change	Admin is authenticated; target account exists.	Open role management; select account; choose new role; confirm.	Target user + Analyst/Admin role	Role is updated and the change is logged.	Not Executed	Not Executed
13_ST_02	User Role Management	Prevent downgrade of last Admin	The selected Admin is the last Admin account.	Attempt to downgrade the last Admin to Analyst.	Last Admin + Analyst role	System denies the request and does not remove the final Admin role.	Not Executed	Not Executed

Manual Testing Record

Before submission, execute each test case against the implemented Web-Based ML Model Evaluator. Replace 'Not Executed' in Actual Result with the observed behavior and mark Test Result as Pass or Fail according to whether the actual result matches the expected result.